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Higher level of communication between the medical staff and the performance staff is associated with a lower hamstring injury burden: a substudy on 14 teams from the UEFA Elite Club Injury Study

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ABSTRACT

Objectives External risk factors connected to club, team and coaching are believed to be important in the causation of hamstring injuries, but little is known about the preventive measures used. The objective was to analyse the association between preventive factors and hamstring muscle injury burden.

Methods 14 teams participated in the Union of European Football Associations Champions/Europa Leagues from 2019/2020 to 2022/2023. An open question was sent out in January 2023 to the four teams with the lowest hamstring muscle injury burden, asking why their teams had successfully avoided hamstring muscle injuries and which preventive methods they had used. A questionnaire that investigated the use of these methods was sent to all 14 teams in February 2023. A multiple linear regression model was applied, using injury burden as an outcome variable and the questionnaire responses as possible explanatory variables possibly adjusted for team and season.

Results The preventive methods included factors such as the level of communication between the coach, the medical staff and the performance staff. The adjusted multiple linear regression model indicated that communication between the medical staff and the performance staff was negatively associated with an increased hamstring muscle injury burden, where for every 'higher' grade of the communication variable, the average injury burden decreased by 4.1 (95% CI −8.2 to −0.04, p=0.048) days per 1000 hours.

Conclusions Better communication between the medical and the performance staff and continuous involvement of the medical staff in load management was associated with a lower hamstring injury burden in male professional football teams.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Hamstring muscle injury is the most frequent and burdensome injury in male professional football players.

⇒ External risk factors connected to clubs, teams and coaching are believed to be of major importance.

⇒ Little is known about the preventive measures used in teams with a lower hamstring muscle injury burden.

INTRODUCTION

The Union of European Football Associations (UEFA) initiated a research project to prevent injuries and increase player safety in male professional football players in 1999—the Elite Club Injury Study (ECIS).[1–6] A major finding in this ongoing observational cohort study with prospectively collected exposure and injury data since 2001 is that hamstring muscle injury consistently has been the most common injury,[1–3] representing, on average, 19% of all reported injuries. This long-term injury surveillance has also provided trend analyses,[1 2 5–8] showing no major reduction in the injury burden of hamstring muscle injuries.

The most recent analysis shows that the incidence and burden of hamstring muscle injuries during both training and match play have increased significantly since 2014/2015 and that the proportion increased from 12% in 2001/2002 to 24% in 2021/2022.[2]

The second step in the sequence of injury prevention is to identify aetiological risk factors behind injuries.[9] In a separate study of perceived risk factors for hamstring muscle injuries in the ECIS, the teams' chief medical officers (CMOs) considered most risk factors associated with external factors connected to

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WHAT THIS STUDY ADDS

⇒ In a search for excellence approach, the chief medical officers of teams with the lowest hamstring muscle injury burden shared their insights on their success in this area.

⇒ Preventive factors identified by these successful teams included club-related aspects such as effective communication among the coach, medical staff and performance staff and active participation in load management. The key findings indicated that improved communication between medical and performance staff and the ongoing engagement of the medical team in load management were linked to a reduced burden of hamstring muscle injuries.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study combines injury surveillance data collected on male professional football players over four consecutive seasons with expertise and knowledge from the chief medical officers of the participating teams.

⇒ This study extends previous findings on the importance of proper communication channels in the team behind the team. Establishing effective communication and cooperation between the medical and performance staff in load management could help clubs reduce the burden of hamstring muscle injury.

⇒ This study highlights the importance of including end-users in the field (such as coaches, medical staff, players, chief executive officers, board members) and incorporating their expertise and knowledge when carrying out risk factor studies at the top-flight level of professional football.

club, team and coaching—all factors that coaches can influence.[10] However, little is known about the preventive measures used in teams with a lower hamstring muscle injury burden.

Therefore, the objective of this study was to analyse the association between the preventive methods used in top-flight football teams and their hamstring muscle injury burden during the 2019/2020 to 2022/2023 seasons of ECIS.

MATERIAL AND METHODS

For this substudy, we invited all teams that had participated in ECIS and qualified for the group stage of either the UEFA Champions League (UCL) or Europa League (EL) in the seasons 2019/2020 to 2022/2023. In total, 14 teams met this criterion and delivered complete injury data for all 4 seasons, and all agreed to answer a substudy questionnaire (response rate 100%). The teams represented eight countries, including three teams from England, three from Germany, two from the Netherlands, two from Spain and one team each from France, Portugal, Belgium and Italy. All teams followed the traditional Western European season. The preseason training period started in July, and the in-season started in August, ending in May. The four seasons' exposure and injury data collection were completed in October 2023.

General methodology

Definitions and data collection methods in ECIS have previously been described in detail.[11–14] In brief, each team selected a medical staff member as the contact person responsible for data registration. Contact persons were given a manual that provided methods and definitions used. Injury was defined as "any physical complaint sustained by a player resulting from a football match or football training that leads to the player being unable to take full part in football training or match play thereafter". A hamstring muscle injury was A traumatic distraction or gradual onset injury to the hamstring muscle group". Hamstring muscle injury burden is expressed as the number of days lost per 1000 hours. Members of the study group reviewed all data to ensure they were complete and complied with the study protocol. If any missing or unclear data were identified during this review, immediate feedback was sent to the contact person to complete or correct the data.

The survey questionnaires

Step 1: developing the search for excellence questionnaire on preventive methods as suggested by the four teams with the lowest hamstring muscle injury burden during the latest four seasons

In January 2023, an open question was sent to the CMOs of the four ECIS teams with the lowest hamstring muscle injury burden during the four seasons 2019/2020, 2020/2021, 2021/2022 and the ongoing season 2022/2023. They were asked to share their opinions on why their teams had successfully avoided hamstring injuries and which preventive methods they had used. The open question was: "*Your team has been more successful than the ECIS average in avoiding hamstring muscle injuries during the latest seasons. Why do you think your team has had fewer hamstring muscle injuries during the three recent seasons?*". The opinions of the four club CMOs, based on experience and theoretical information, are presented in box 1.

Step 2: questionnaire on the practical use of the suggested preventive methods

A questionnaire investigated the practical use of the six preventive factors suggested by the four teams with the lowest hamstring muscle injury burden during the four

Box 1 Preventive methods used by the four teams that had the lowest hamstring muscle injury burden during the four seasons 2019/2020, 2020/2021, 2021/2022 and 2022/2023

1. Established good communication between medical staff and head coach (4/4 teams).
2. Established good communication between medical staff and performance staff (4/4 teams).
3. Established an optimal load on players and team.
4. Had medical staff involved in load management (3/4 teams).
5. Had monitoring of regular exposure to high-speed football actions during training (3/4 teams).
6. Had individualised preactivation regimes based on each player's weaknesses (3/4 teams).
7. Had training/exercise surveillance by head coach (1/4 teams).

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Box 2 Questionnaire on the practical use of the suggested preventive methods

1. How would you rate the level of communication between the medical staff and the head coach?
   (On a scale from 1 to 10, where 1 is the worst possible and 10 is the best possible)
2. How would you rate the level of communication between the medical staff and performance staff?
   (On a scale from 1 to 10, where 1 is the worst possible and 10 is the best possible)
3. Were you involved in the load management?
   (Yes, no; other, please specify)
4. Established an optimal load on players and team
5. Was regular exposure to high-speed football actions at training monitored?
   (Yes, no; comments)
6. Did you have an individualised preactivation regimen based on each player's weaknesses?
   (Yes, no; comments)
7. Was the head coach involved in surveilling all training activities, including prevention training?
   (Yes, no; comments)

latest seasons (box 2). Two questions provided a 10-point Numerical Rating Scale; four had a dichotomous yes/no response. This questionnaire was distributed to the CMOs of all 14 teams in February 2023 during the ongoing 2022–2023 season. The questionnaire covered all four seasons and was separated for each season. After agreeing to participate in the survey, the CMOs were provided access to the questionnaire using the online survey software SurveyMonkey (SurveyMonkey, California, USA). The survey software distributed automatic reminders after three, seven and ten days. All 14 teams completed the questionnaire within a month and provided data for four seasons each (ie, in total 56 observations).

Missing data

One club did not provide survey data in the 2019/2020 season questionnaire, and another team did not provide questionnaire data for the 2021/2022 season, reducing the number of observations to 54.

Patient and public involvement

This research was carried out without patient (player) involvement. Players were not invited to comment on the study design or contribute to drafting this document.

Statistical analysis

The hamstring muscle injury burden, calculated as the number of absence days per 1000 hours, was used as an outcome measure. A linear regression model with stepwise backward elimination (ie, excluding the non-significant explanatory variables stepwise) was applied with injury burden as an outcome variable and the following as possible explanatory variables: established good communication between medical staff and head coach (scale 1–10), established good communication between medical staff and performance staff (scale 1–10), had medical staff involved in load management (0, no; 1, yes), had regular exposure to and monitoring of high-speed football actions during training (0, no; 1, yes), had individualised preactivation regimen based on each player's weaknesses (0, no; 1, yes) and had training/exercise surveillance by the head coach (0, no; 1, yes). The variables team (categorical, 1–14 teams) and season (categorical, 1–4 seasons) were used as adjustment variables in the analyses if significant. Associations with $p<0.05$ were considered statistically significant, and the coefficient of determination (R^2) is presented. The analyses used the Statistical Package for the Social Sciences (SPSS) for Windows V.29.0. Because the four teams with the lowest hamstring muscle injury burden that provided the preventive factors asked about in the survey are included in the analysis, there is a risk that the results may be biased. Therefore, a sensitivity analysis is performed, and these four teams are excluded from the analysis.

RESULTS

Figure 1 shows the mean and seasonal hamstring muscle injury burden over the four seasons for each of the 14 teams; it illustrates the large between-team and within-team differences, with the mean injury burden over the four seasons varying from 9 to 48 days absence per 1000 hours. Figure 1 also illustrates the rationale behind selecting the four teams approached by the 'search of excellence' questionnaire (box 1). Teams 1–4 were selected since they had the lowest hamstring muscle injury burden sums over the four seasons, varying between 9 and 20 days of absence per 1000 hours per season.

Table 1 shows the burden of hamstring muscle injury and survey responses to the questionnaire on the practical use of the suggested preventive methods (box 2). The four excellence teams had a lower-than-average hamstring injury burden, below the seasonal mean day's absence in 15 out of 16 team seasons, with only team 4 in the 2022/2023 season having 33 days of absence per 1000 hours, compared with the seasonal mean of 31 days of absence per 1000 hours. All teams except one experienced at least one out of the four seasons where their hamstring injury burden was lower than the seasonal mean.

Association between injury burden and predictors

The median (IQR) levels of communication between the medical staff and the head coach and the communication between the medical staff and performance staff were on the same level: 8 (2) and 8 (3).

Regarding preventive actions, an individualised preactivation regimen based on each player's weaknesses was implemented in 50 of the 54 team seasons (93%), regular exposure to high-speed football actions at training was monitored in 48 team seasons (89%), the medical staff was involved in load

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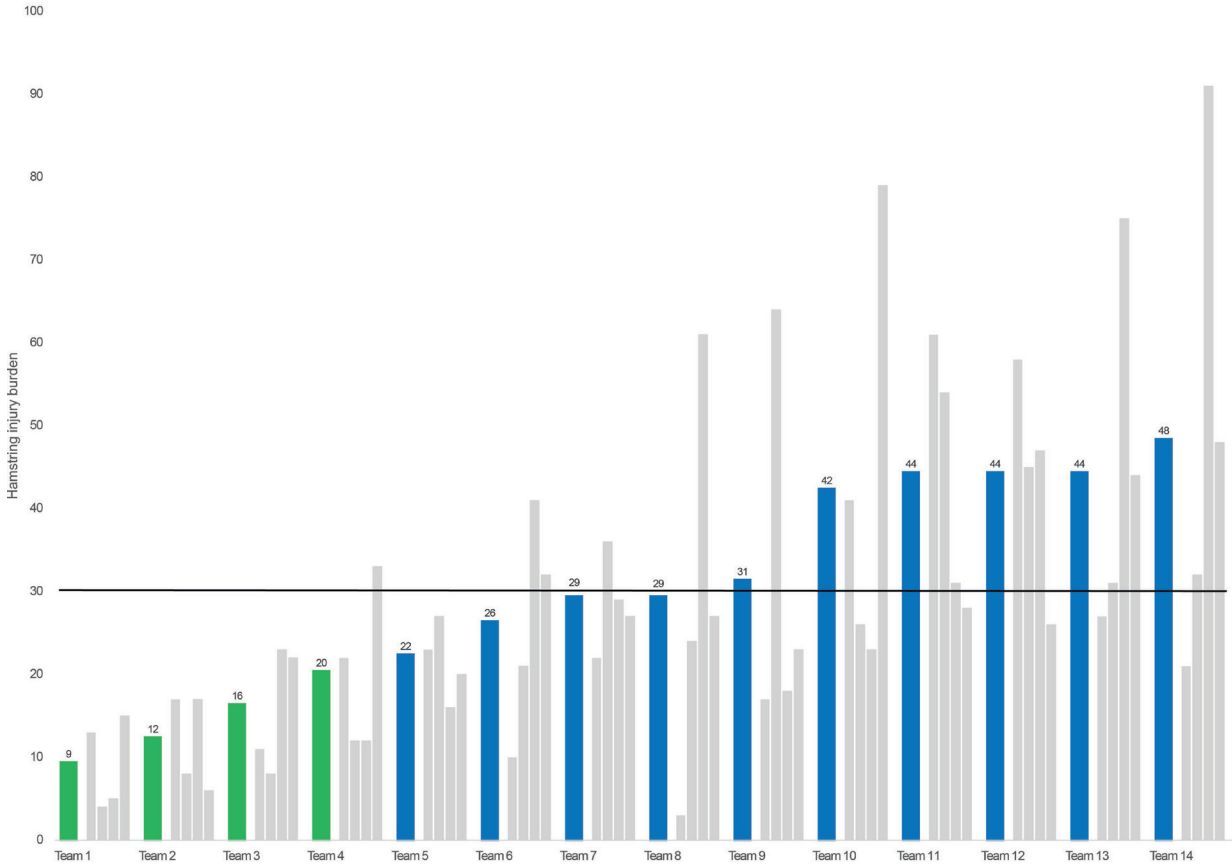

Figure 1 The mean and actual hamstring muscle injury burden per team and season. The green and blue bars illustrate the mean injury burden per team expressed as the mean number of days of absence per 1000 hours over the four seasons. Green bars represent the four excellence teams, and blue bars represent the other ten. Grey bars illustrate each season (each team's first and last bar shows seasons 2019/2020 and 2022/2023, respectively). The horizontal black line represents the mean over all 4 seasons for all 14 teams.

management in 34 team seasons (63%) and the head coach was involved in the surveillance of all training activity in 28 team seasons (52%).

The adjusted multiple linear regression model (R^2=0.33) indicated that communication between the medical staff and the performance staff was negatively associated with an increased hamstring muscle injury burden, where for every 'higher' grade of the communication variable, the average injury burden decreased by 4.1 (95% CI −8.2 to −0.04, p=0.048) days per 1000 hours (table 2). Additionally, if the medical staff were involved in the load management, the average injury burden increased by 18 (95% CI 1.9 to 35, p=0.030) days per 1000 hours (table 2). These two variables were also found to be significant in the simple linear regression model, where conversely, involvement of the medical staff in load management had a negative association with an increased hamstring muscle injury burden, meaning that the hamstring muscle injury burden decreased by 12 days per 1000 hours in seasons with medical staff involvement. Notably, teams where the medical staff was consistently involved in load management during all observed seasons (table 1, question 3; teams #1, 2, 3, 4, 7 and 9), had a lower injury burden than teams where the medical staff was not involved at all or had inconsistent involvement over the observed seasons (teams #5, 6, 8, 10, 11, 12, 13 and 14), with an average season burden of 19.6 vs 37.3 days per 1000 hours.

The results from the sensitivity analysis, when the four teams that provided preventive factors were excluded, confirmed the results of the main analysis; the β-coefficients of communication between the medical staff and the performance staff and the medical staff involved in the load management were of similar size as in the main analysis (−4.4 and 18.7 compared with −4.1 and 18.3).

DISCUSSION

The main finding was that a higher level of communication between the medical and performance staff was associated with a lower hamstring muscle injury burden.

Further relevant findings were as follows:

- There was a large within-season variation of the hamstring muscle injury burden between the 14 teams studied.
- There was considerable variation in hamstring muscle injury burden across different seasons for the same teams.

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Table 1 Hamstring muscle injury burden and survey responses per team and season

Team	Season	Injury burden (days/1000 hours)	The survey question (box 2) and individual team response					
			1	2	3	4	5	6
#1	2019/2020	13	–	–	–	–	–	–
	2020/2021	4	8	7	Yes	Yes	Yes	No
	2021/2022	5	7	8	Yes	Yes	Yes	Yes
	2022/2023	15	6	7	Yes	Yes	Yes	No
#2	2019/2020	17	7	9	Yes	Yes	Yes	No
	2020/2021	8	7	9	Yes	Yes	Yes	No
	2021/2022	17	7	9	Yes	Yes	Yes	No
	2022/2023	6	9	9	Yes	Yes	Yes	No
#3	2019/2020	11	8	10	Yes	Yes	Yes	Yes
	2020/2021	8	8	10	Yes	Yes	Yes	Yes
	2021/2022	23	8	10	Yes	Yes	Yes	Yes
	2022/2023	22	8	10	Yes	Yes	Yes	Yes
#4	2019/2020	22	10	9	Yes	Yes	Yes	Yes
	2020/2021	12	10	9	Yes	Yes	Yes	Yes
	2021/2022	12	10	9	Yes	Yes	Yes	Yes
	2022/2023	33	9	10	Yes	Yes	Yes	Yes
#5	2019/2020	23	8	6	No	Yes	Yes	No
	2020/2021	27	8	7	Yes	Yes	Yes	Yes
	2021/2022	16	8	10	Yes	Yes	Yes	Yes
	2022/2023	20	7	9	Yes	Yes	Yes	No
#6	2019/2020	10	7	7	No	Yes	Yes	No
	2020/2021	21	7	6	No	No	Yes	No
	2021/2022	41	9	8	Yes	Yes	Yes	No
	2022/2023	32	8	6	No	Yes	Yes	No
#7	2019/2020	22	10	10	Yes	Yes	Yes	Yes
	2020/2021	36	10	10	Yes	Yes	Yes	Yes
	2021/2022	29	10	10	Yes	Yes	Yes	Yes
	2022/2023	27	10	10	Yes	Yes	Yes	Yes
#8	2019/2020	3	7	8	Yes	No	Yes	No
	2020/2021	24	7	5	Yes	Yes	No	No
	2021/2022	61	7	3	No	No	No	No
	2022/2023	27	8	9	Yes	Yes	Yes	Yes
#9	2019/2020	17	10	9	Yes	Yes	Yes	No
	2020/2021	64	9	9	Yes	Yes	Yes	No
	2021/2022	18	9	9	Yes	Yes	Yes	No
	2022/2023	23	9	9	Yes	Yes	Yes	Yes
#10	2019/2020	41	10	8	No	Yes	Yes	Yes
	2020/2021	26	5	4	No	Yes	Yes	Yes
	2021/2022	23	–	–	–	–	–	–
	2022/2023	79	6	6	Yes	Yes	Yes	No
#11	2019/2020	61	8	3	No	No	No	No
	2020/2021	54	9	9	No	Yes	Yes	Yes
	2021/2022	31	9	9	No	Yes	Yes	Yes
	2022/2023	28	3	9	No	No	No	No

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Table 1 Continued

Team	Season	Injury burden (days/1000 hours)	The survey question (box 2) and individual team response					
			1	2	3	4	5	6
#12	2019/2020	58	6	6	No	Yes	Yes	No
	2020/2021	45	7	9	Yes	Yes	Yes	Yes
	2021/2022	47	7	7	No	Yes	Yes	No
	2022/2023	26	7	7	No	Yes	Yes	No
#13	2019/2020	27	5	6	No	Yes	Yes	Yes
	2020/2021	31	5	6	No	Yes	Yes	Yes
	2021/2022	**75**	5	6	No	Yes	Yes	Yes
	2022/2023	44	6	5	No	Yes	Yes	Yes
#14	2019/2020	21	5	6	No	No	Yes	No
	2020/2021	32	6	8	No	Yes	Yes	No
	2021/2022	91	10	8	Yes	Yes	Yes	Yes
	2022/2023	48	10	7	Yes	Yes	Yes	Yes

Red cells indicate a below-median survey response, green cells indicate a yes or above median response or above (red)/below (green) mean hamstring injury burden. Grey cells indicate no response.

- The injury burden related to hamstring muscles was lower when medical staff played a consistent role in load management; conversely, in teams where their involvement fluctuated between seasons, the injury burden was generally higher.
- There was no difference between the four 'excellence teams' and the other ten teams regarding individualised preactivation regimens based on each player's weaknesses. In addition, they all had regular exposure to high-speed football actions during training, and the head coach was surveilling all training activities.

Why was a higher grade of communication between the medical and performance staff associated with a lower injury burden?

A previous study from ECIS (2012–2016) found that good internal communication within teams led to lower injury rates, increased training attendance and improved match availability.[15] Effective communication between the head coach and medical staff resulted in fewer injuries, whereas poor communication between medical staff and fitness coaches lowered player training attendance. During recent years, however, the role of the head coach in top-flight football teams has evolved to be much like that of an orchestra conductor.[16 17] The modern head coach must show diplomatic communication skills and be able to delegate effectively among the team behind the team.[16 17] The head of fitness/performance now significantly influences training loads within the coaching framework.[16 17] Given that player loading and fatigue are linked to hamstring muscle injuries, the fitness/performance coach may directly impact injury rates more than the head coach. However, the head coach still guides the overall team strategy. The medical staff plays a crucial role in planning physical

Table 2 Determinants for hamstring muscle injury burden obtained by simple and multiple linear regression models

Explanatory variable	Simple, β-coefficient (95% CI) (p value)	Multiple, β-coefficient (95% CI) (p value)
Level of communication between medical staff and head coach	−0.4 (−3.7 to 2.8) (0.769)	
Level of communication between medical staff and performance staff	−4.1 (−6.8 to −1.3) (0.004)	−4.1 (−8.2 to −0.04) (0.048)
Medical staff involved in load management	−12 (−23 to −1.0) (0.033)	18 (1.9 to 35) (0.030)
Head coach involved in the surveillance of all training activity	2.1 (−8.8 to 13) (0.707)	
Monitoring of regular exposure to high-speed football actions at training	−2.7 (−20 to 15) (0.753)	
Individualised preactivation regimen based on each player's weaknesses	−14 (−35 to 5.9) (0.161)	

Regression coefficients in the simple linear and the multiple linear regression models. Non-significant variables (ie, p>0.05) were excluded stepwise from the multiple models. The variable team is included in the multiple models as an adjusted variable that is significant but not seasonal.

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loads and managing injury treatments and return-to-play discussions.

What are the practical consequences of variation in injury burden between teams?

- **Impact on success:** Winning matches and trophies is the main aim in professional football, which depends on player availability for training and games. Typically, 12%–14% of UCL players are unavailable due to injury, with individual team rates ranging from 4% to 23%.[6 18] Lower injury rates correlate with greater success in competitions.[19 20]
- **Financial implications:** Injuries in a UCL starting eleven can cost clubs €17 000–€20 000 daily.[18 21] With UCL squads facing around 1100 days of player absences each season, the annual injury-related costs can reach about €20 million, highlighting the significant economic impact on club management.[18 19]

Why was there a within-team variation in injury burden between seasons?

Notably, there was significant variation in hamstring injury muscle burden between teams within a season and across different seasons. This emphasises the need for studies to involve multiple teams across several seasons to enhance the validity of findings.[12–14] In 'one-club, one-season' injury studies, whether the season analysed is typical or an anomaly can be unclear. Several factors may contribute to the variation within teams, such as changes in squad composition, including adding more injury-prone players (eg, those with past hamstring muscle injuries or those who engage in high-speed running). Additionally, shifts in coaching and performance staff can affect injury rates, as any role changes can alter group dynamics and communication, which are crucial in this context.[16]

Can we clarify the conflicting findings regarding the link between medical staff involvement in load management and injury burden?

The burden of hamstring muscle injuries was lower when medical staff consistently engaged in load management. In contrast, teams where medical staff involvement varied from season to season experienced a higher injury burden. Our investigation focused on teams with inconsistent medical staff participation across seasons—where the medical team was involved in some seasons but not others—and indicated that the average injury burden increased during seasons when medical staff contributed to load management. However, the injury burden decreased when medical staff were consistently involved in load management. Inconsistent practices led to a higher injury burden during seasons when medical staff were active.

This paradox may not stem from a lack of expertise in load management but rather from inconsistent communication and collaboration, with medical staff participating sporadically. As noted in the ECIS, effective internal team communication correlates with injury rates and player availability.[15] Establishing a structured and consistent communication framework among performance and technical staff can promote a unified understanding of load management.

Another potential explanation is that medical staff involvement may fluctuate depending on the team's injury status; for instance, they might engage when there is a notable increase in hamstring muscle injuries to 'address the issue'. Additionally, one could speculate that some clubs maintain a philosophy of always involving the medical team in load management, independent of coaching changes. In contrast, other teams leave the decision to the head coach.

Is implementation the problem?

All medical staff recognise the necessity of preventive actions, but their execution may rely on the specific club situation and the support from the executive board and head coach. The effectiveness of individual screenings for weaknesses also varies. For preventive measures to succeed, players, coaches and officials must be highly motivated to engage with the programme.[22–25] Research on professional athletes shows challenges in applying interventions in controlled versus clinical settings.[23 25] Implementation requires teamwork, and gaining buy-in from players and club staff can take time.

How could these findings impact future clinical practice?

Effective communication is vital, but increasing support staff complicates this. A structured communication strategy among all stakeholders is recommended. Clubs should unify staff roles, define responsibilities and ensure a transparent decision-making process to aid coaching. Additionally, further research on how medical decision-making affects club performance, both on and off the field, is essential for validation.

Methodological considerations

Hamstring muscle injury data were collected prospectively, providing robust insights due to the 21-year history of the ECIS. This tool is reliable for assessing injury risk and patterns in male professional footballers, with a 100% response rate from all 14 teams. However, the study has limitations. Team doctors provided survey responses for four seasons in the 2022/2023 season, leading to potential recall bias for earlier seasons. Additionally, the inclusion criteria encompassed a range of hamstring muscle injuries, from structural (partial or total ruptures) to functional (no visible disruption), with previous data showing that 71% were structural.[2] The study did not explore the distinct causes of acute versus gradual injuries. Furthermore, the findings may not be generalisable to other professional contexts. Causality between expert preventive opinions and injury burden cannot be established due to varying risk factors among clubs. As this is an ecological study, further research with a better design and larger samples is necessary. Finally, differences in

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match play and fixture congestion across seasons could affect injury rates.[26–28] However, this substudy focused only on teams that competed in the UCL or EL across all four seasons.

CONCLUSIONS

A lower hamstring muscle injury burden was associated with a higher level of communication between the medical staff and the performance staff. Successful teams with low injury rates emphasised club-related factors, including good communication between the coach, medical and performance staff and ongoing medical involvement in load management. These factors warrant further investigation in prospective studies.

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